A COUNTABLY GENERATED MITTAG-LEFFLER MODULE OVER A HENSELIAN LOCAL RING HAS NO ASSOCIATED PRIME OF A TENSOR PRODUCT MEETING THE EXTENDED MAXIMAL IDEAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Splitting of a countably generated Mittag-Leffler module over a henselian local ring as a direct sum of finite modules. Algebra, label algebra-lemma-split-ML-henselian (invoked at proof line 4432). It reduces the general module to the case of a direct sum of finite modules, where the conclusion is known.
- The corresponding statement for a direct sum of finite modules. Same chapter flat.tex, label lemma-explain-why-pure-direct-sum-finite-modules (invoked at proof line 4430). It is the case to which the splitting reduces, and it supplies the conclusion about associated primes of $N \otimes_R M$.

Source reference. Original lemma 05D6: sources/stacks-chow/tags.txt, source lines 6944–6944. Navigation pointer for \ref{algebra-lemma-split-ML-henselian} used inside the counted span (line 4432); the argument that consumes it is retained in the span.

Source reference. Original lemma 05IU: sources/stacks-flat/pinned-original-source.tex, source lines 4379–4379. Navigation pointer for \ref{lemma-explain-why-pure-direct-sum-finite-modules} used inside the counted span (line 4430); the argument that consumes it is retained in the span.

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Lemma 0.1. *Let R be a henselian local ring with maximal ideal $\mathfrak{m}$. Let $R \to S$ be a ring map. Let N be an S-module. Assume N is countably generated and Mittag-Leffler as an R-module. Then for any R-module M and for any prime $\mathfrak{q} \subset S$ which is an associated prime of $N \otimes_R M$ we have $\mathfrak{q} + \mathfrak{m}S \neq S$.*

Proof. This lemma reduces to Lemma 05IU by Algebra, Lemma 05D6. □